Electrical Circuit (1)

Linearity circuit (week7 class1)

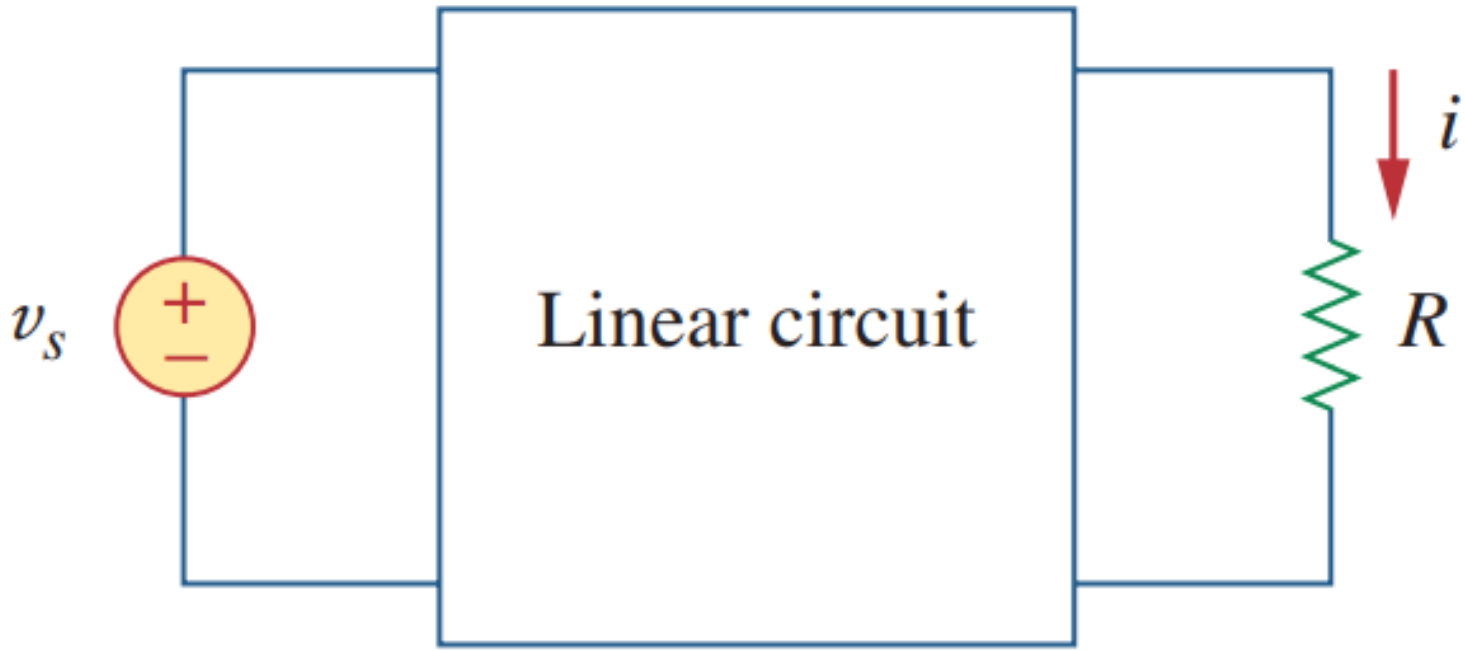
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Linearity Property



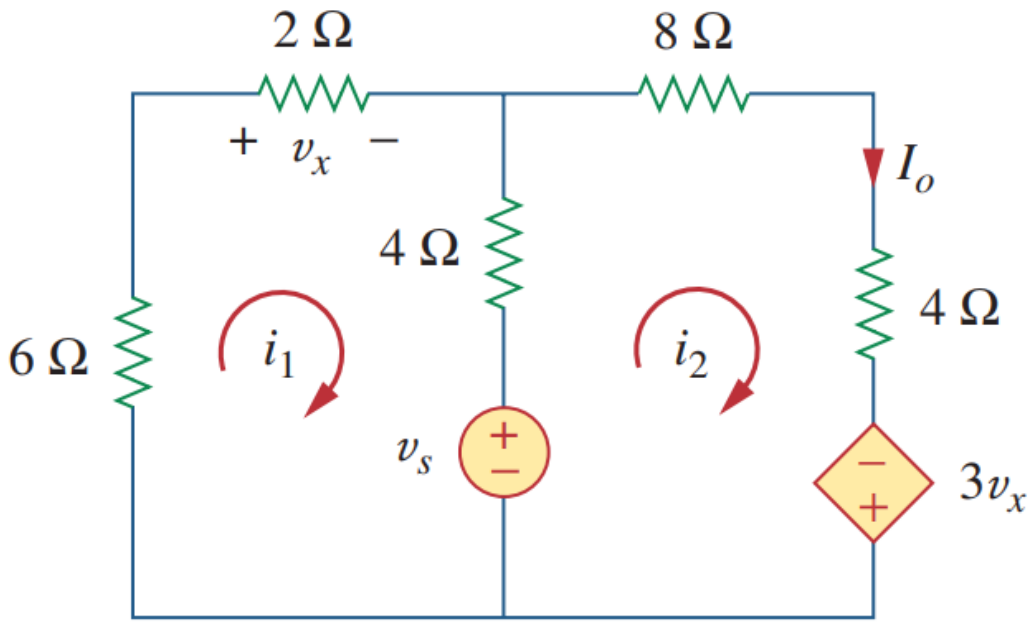
Linearity Property

For the circuit find I_o when $v_s = 12$ V and $v_s = 24$ V.

Basic equation

$$(6 * i_1) + (2 * i_1) + 4 * (i_1 - i_2) + 12 = 0$$

$$-12 + 4 * (i_2 - i_1) + (8 * i_2) + (4 * i_2) - 3(2 * i_1) = 0$$



Formulated equation

$$12i_1 - 4i_2 = -12$$

$$-10i_1 + 16i_2 = 12$$

Final answer @ Vs =12V

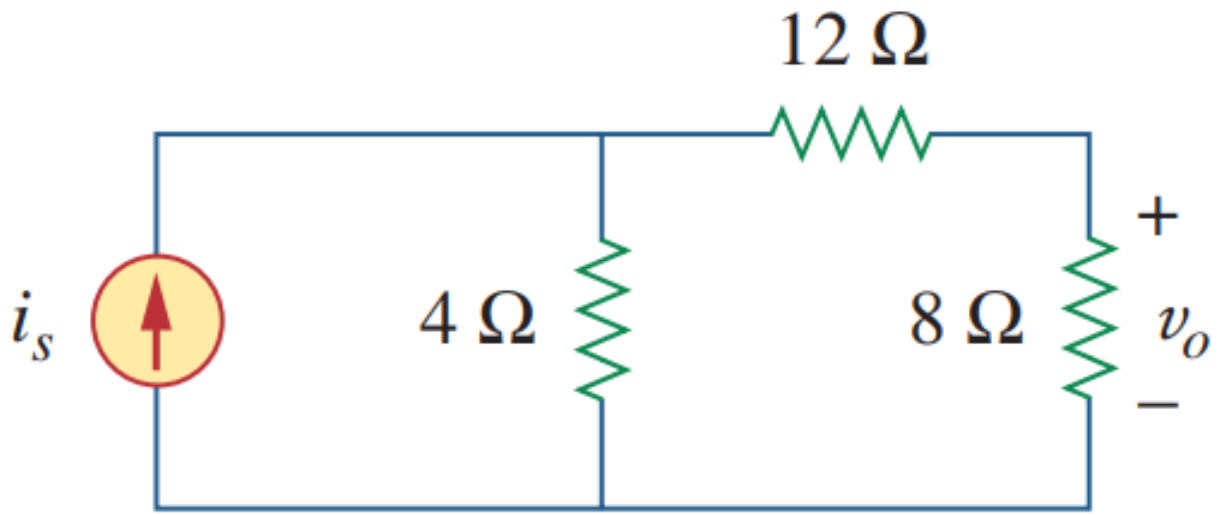
$$i_1 = \frac{-18}{19}\ \text{A} \qquad i_2 = \frac{3}{19} A$$

Final answer @ Vs =24V

$$i_1 = \frac{-32}{19}\ \text{A} \qquad i_2 = \frac{6}{19} A$$

Linearity Property

For the circuit find v_o when $i_s = 30$ and $i_s = 45$ A.



@ $i_s = 30$

$$i_0 = i_s \frac{4}{12+8} = 30\frac{4}{12+8+4} = 5A$$

$$v_0 = 5 * 8 = 40\ V$$

@ $i_s = 45$

$$\frac{i_{s,new}}{i_{s,old}} = \frac{v_{o,new}}{v_{o,old}}$$

$$v_{o,new} = v_{o,old}\frac{i_{s,new}}{i_{s,old}} = 40\frac{45}{30} = 60V$$